

CS675

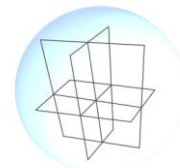
Computer Graphics

Viewing Transformations

Parag Chaudhuri

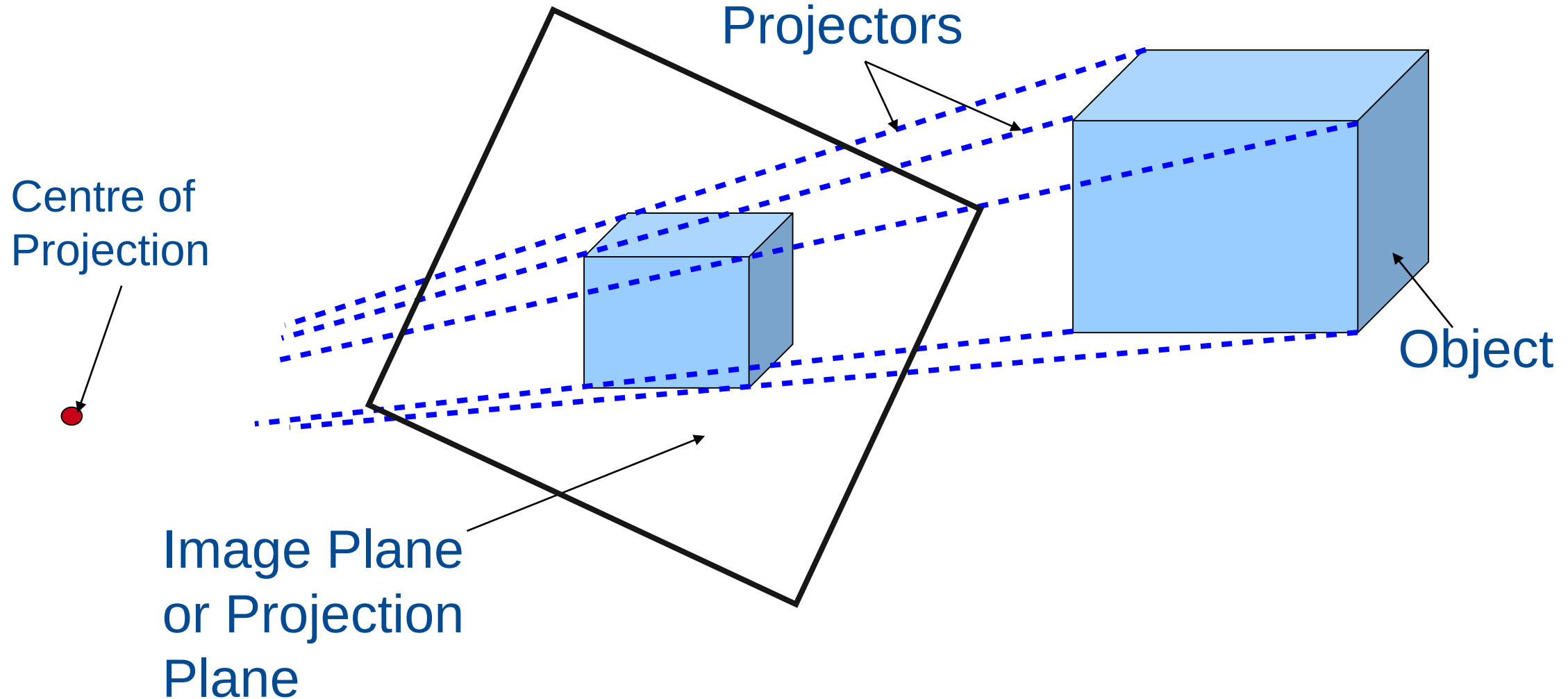


Department of Computer Science and Engineering
IIT Bombay

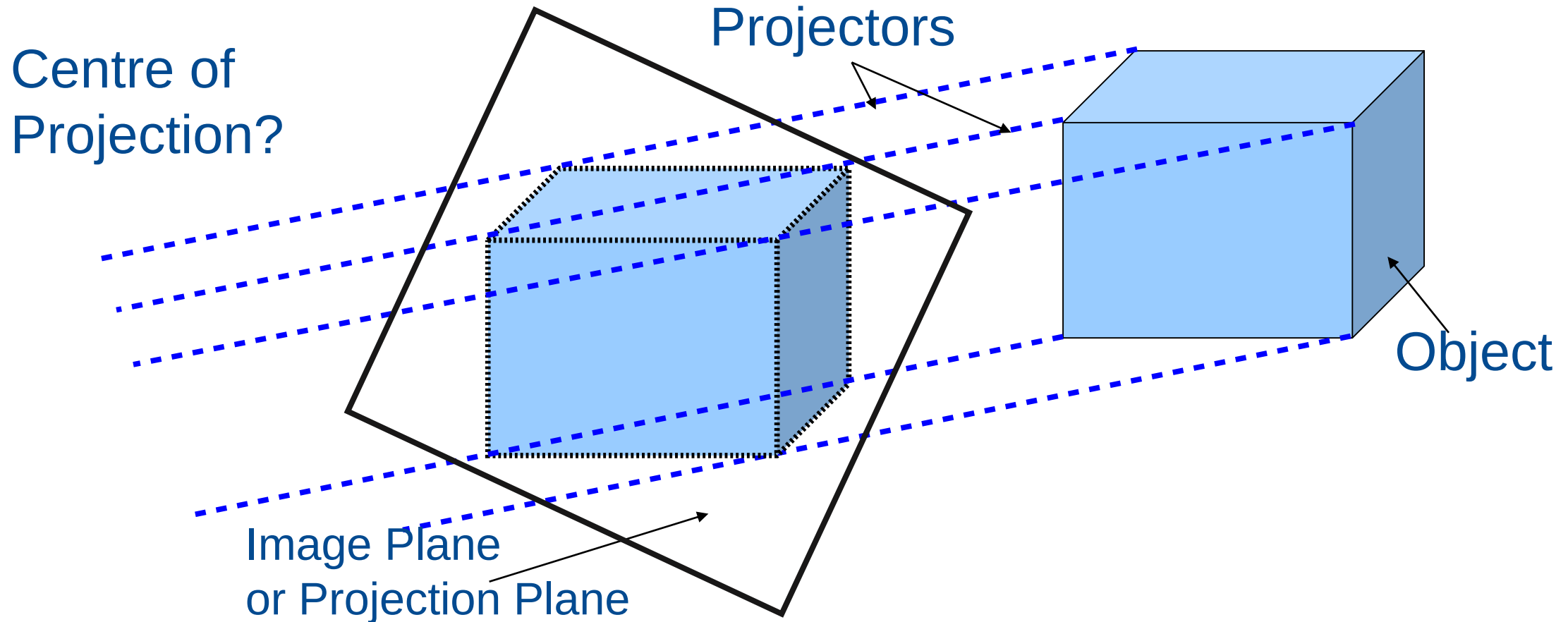


ViGIL
VISION, GRAPHICS,
IMAGING & LEARNING

Perspective Projection



Parallel Projection

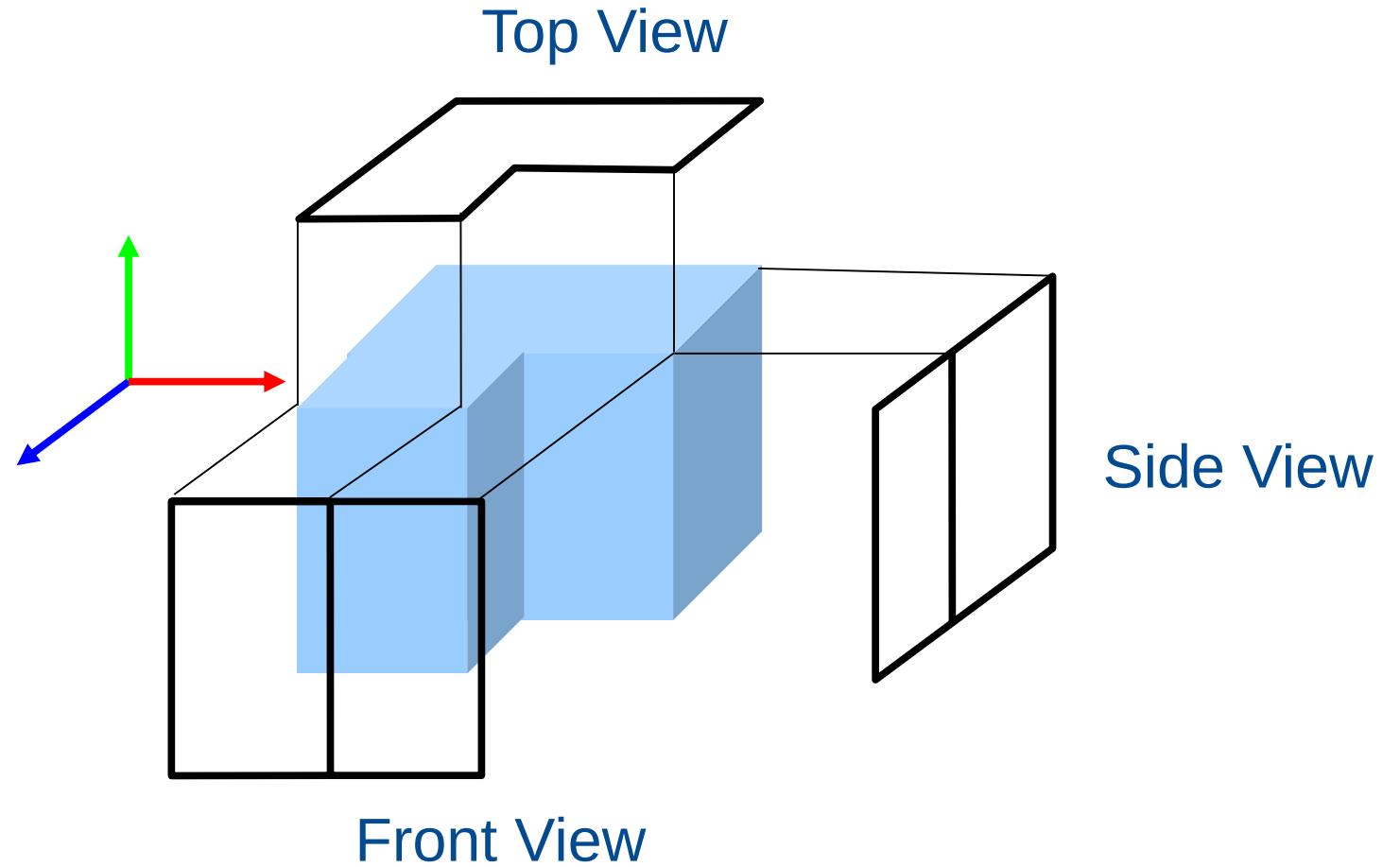


Parallel Projection

Orthographic Projection

- Multiviews ($x=0$, $y=0$ or $z=0$ or principal planes).
- True size or shape for lines.
- For projection on the $z=0$ plane we get the projection matrix as

$$P(z = 0) = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$



Parallel Projections

Parallel Projection

Axonometric Projection

- Transform and then project using an orthographic projection such that at multiple adjacent faces are visible – better representation of a 3D object using 1 view. Face parallel to projection plane shows true shape and size.
- If U be the matrix formed by stacking up the unit vectors along the three axes, and T be the axonometric projection, then

$$T.U = T \cdot \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 1 & 1 \end{bmatrix} = \begin{bmatrix} x_x' & x_y' & x_z' \\ y_x' & y_y' & y_z' \\ 0 & 0 & 0 \\ 1 & 1 & 1 \end{bmatrix}$$

Parallel Projection

Axonometric Projection

- If U be the matrix formed by stacking up the unit vectors along the three axes, and T be the axonometric projection, then

$$T.U = T \cdot \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 1 & 1 \end{bmatrix} = \begin{bmatrix} x_x' & x_y' & x_z' \\ y_x' & y_y' & y_z' \\ 0 & 0 & 0 \\ 1 & 1 & 1 \end{bmatrix}$$

- The foreshortening ratios for each projected principal axes are then given by:

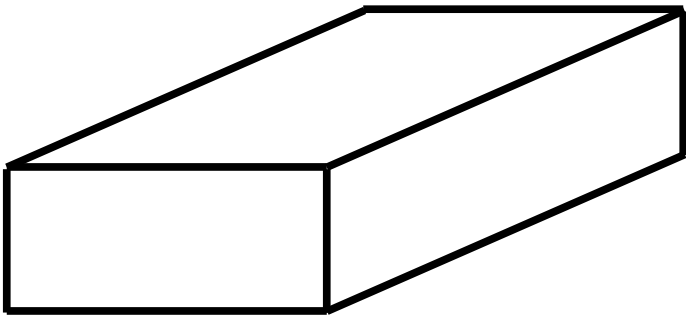
$$f_x = \sqrt[2]{x_x'^2 + y_x'^2} \quad f_y = \sqrt[2]{x_y'^2 + y_y'^2} \quad f_z = \sqrt[2]{x_z'^2 + y_z'^2}$$

Parallel Projection

Axonometric Projection

- Depending on the kind of foreshortening they cause we can have three types of axonometric projections
 - Trimetric (all foreshortenings are different)

$$f_x \neq f_y \neq f_z$$

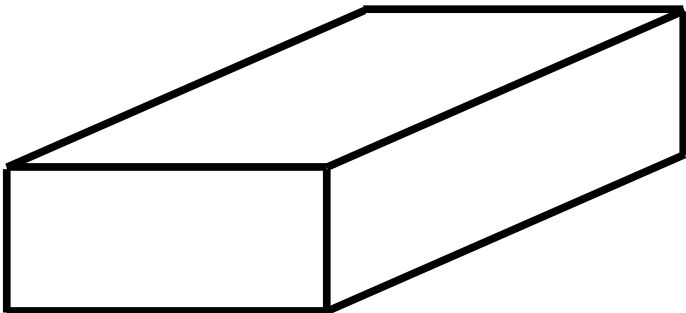


Parallel Projection

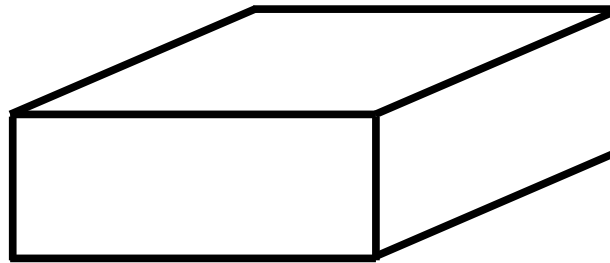
Axonometric Projection

- Depending on the kind of foreshortening they cause we can have three types of axonometric projections
 - Trimetric (all foreshortenings are different)
 - Dimetric (two foreshortenings are the same)

$$f_x \neq f_y \neq f_z$$



$$f_x = f_z$$

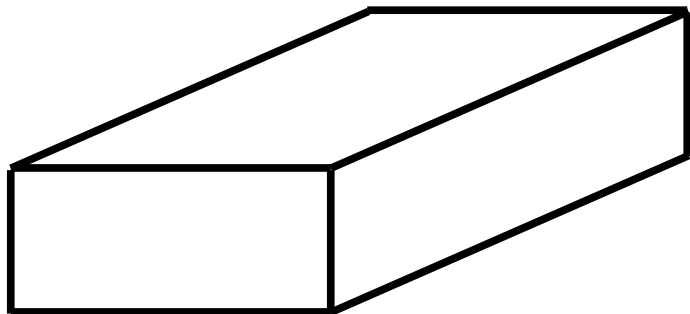


Parallel Projection

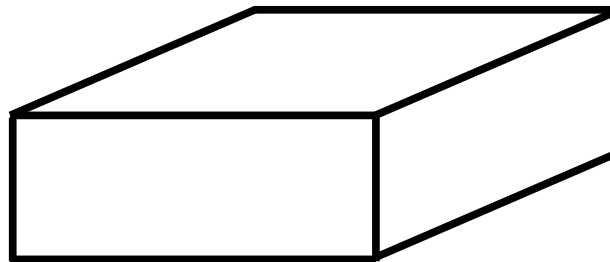
Axonometric Projection

- Depending on the kind of foreshortening they cause we can have three types of axonometric projections
 - Trimetric (all foreshortenings are different)
 - Dimetric (two foreshortenings are the same)
 - Isometric (all foreshortenings are the same)

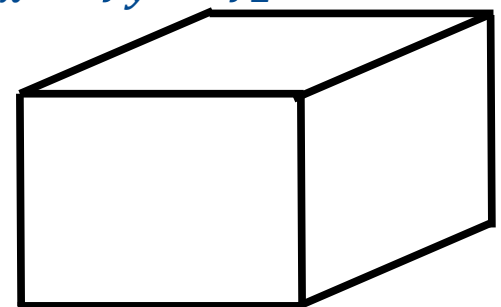
$$f_x \neq f_y \neq f_z$$



$$f_x = f_z$$



$$f_x = f_y = f_z$$



Parallel Projection

Axonometric Projection

- Assuming we rotate by $R_y(\phi)$ and $R_x(\theta)$ before we do the projection on the $z=0$ plane.

$$T = P(z = 0) \cdot R_x(\theta) \cdot R_y(\phi)$$

Parallel Projection

Axonometric Projection

- Now we apply this axonometric projection T to U

$$\begin{aligned} T.U &= \begin{bmatrix} \cos\phi & 0 & \sin\phi & 0 \\ \sin\theta\sin\phi & \cos\theta & -\sin\theta\cos\phi & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 1 & 1 \end{bmatrix} \\ &= \begin{bmatrix} \cos\phi & 0 & \sin\phi \\ \sin\theta\sin\phi & \cos\theta & -\sin\theta\cos\phi \\ 0 & 0 & 0 \\ 1 & 1 & 1 \end{bmatrix} \end{aligned}$$

Parallel Projection

Axonometric Projection

- So the foreshortening ratios become

$$f_x^2 = \cos^2 \phi + \sin^2 \theta \sin^2 \phi$$

$$f_y^2 = \cos^2 \theta$$

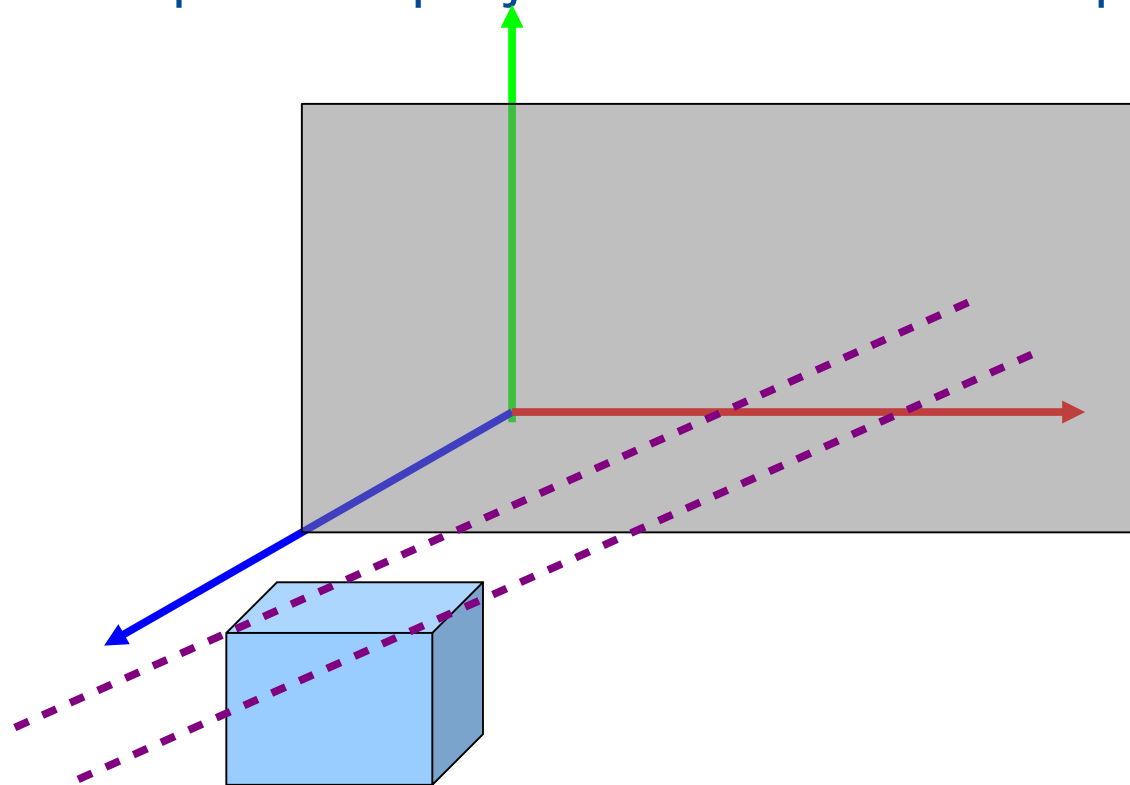
$$f_z^2 = \sin^2 \phi + \sin^2 \theta \cos^2 \phi$$

- For Isometric projections, if we solve for $f_x = f_y = f_z$, then we get $\theta = 35.26^\circ$ and $\phi = \pm 45^\circ$

Parallel Projection

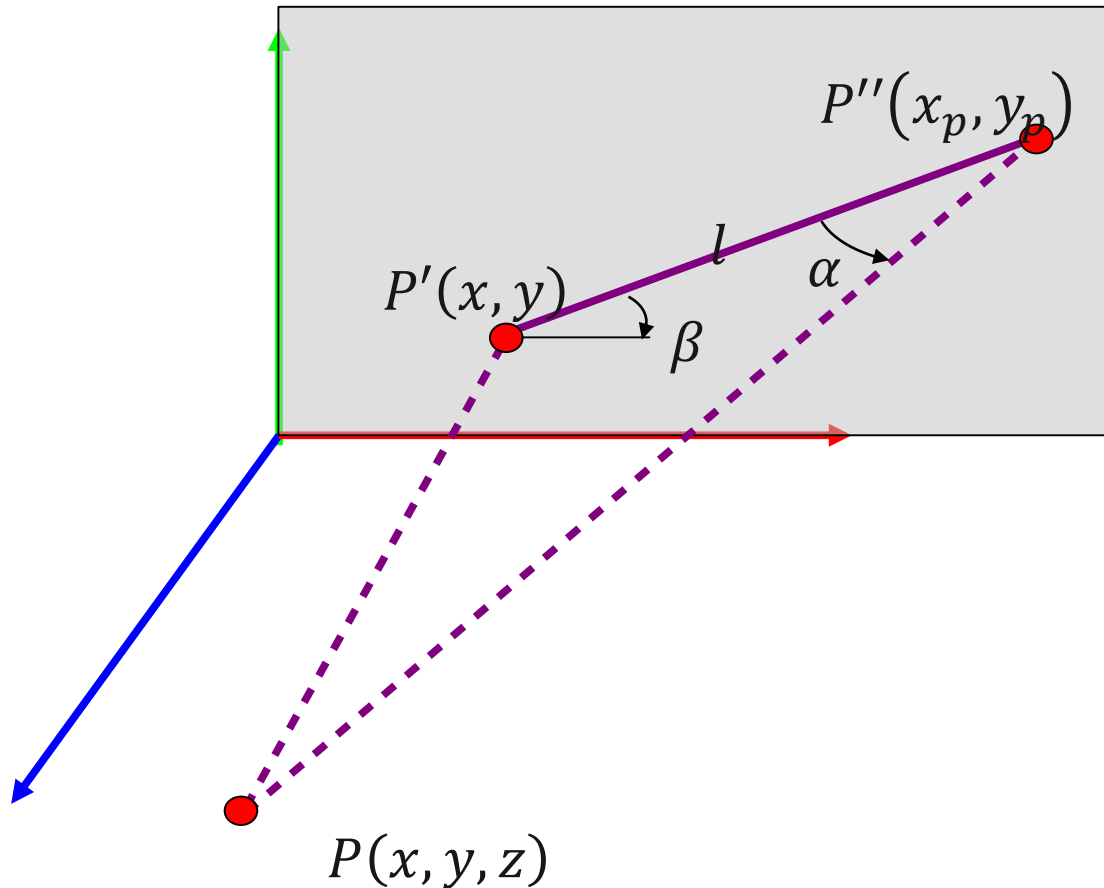
Oblique Projection

- The projectors are parallel to each other but they are not perpendicular to the plane of projection.
- Only planes parallel to plane of projection show true shape and size.



Parallel Projection

Oblique Projection



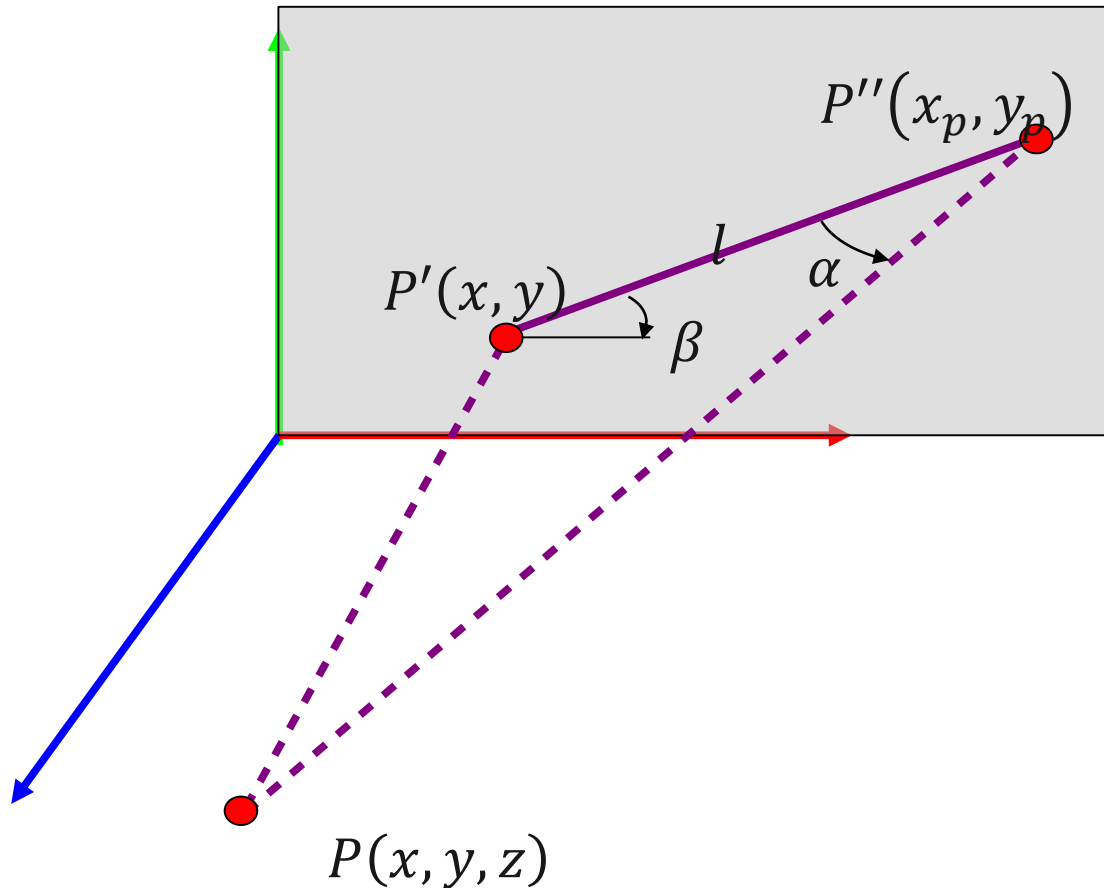
$$x_p = x + l \cos \beta$$

$$y_p = y + l \sin \beta$$

$$\tan \alpha = \frac{z}{l} \quad \text{or} \quad l = z \cot \alpha$$

Parallel Projection

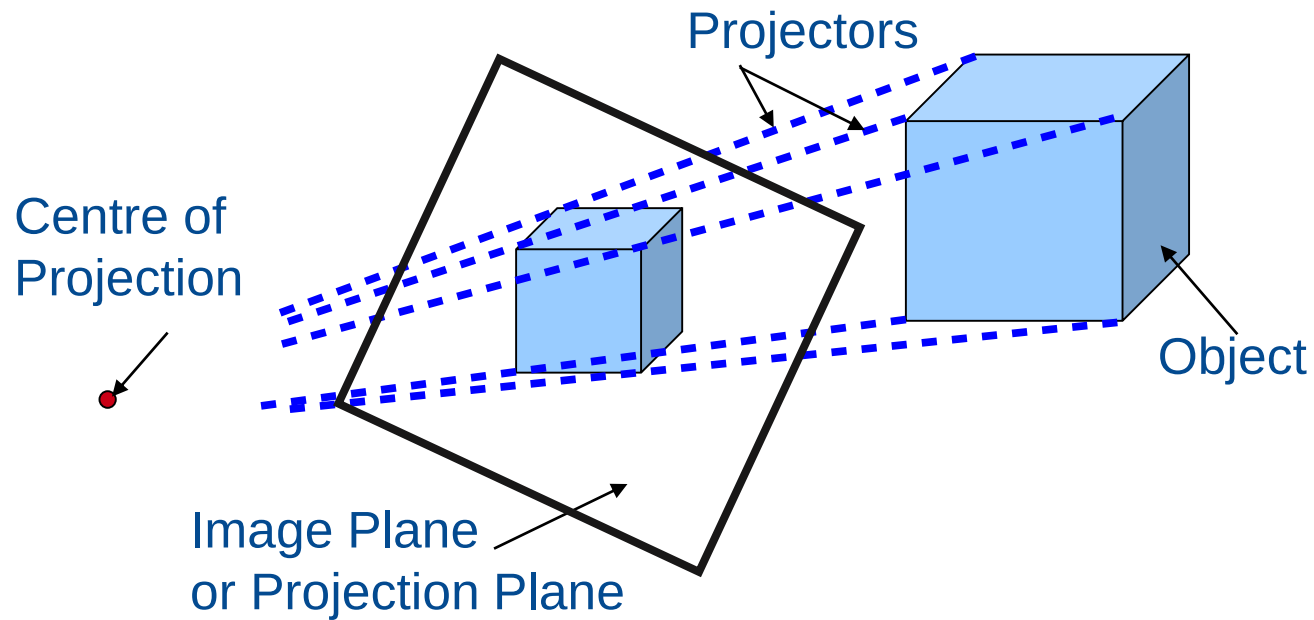
Oblique Projection



- When $\alpha = 45^\circ$ we get a **Cavalier** projection. Lines perpendicular to the projection plane are not foreshortened.
- When $\cot \alpha = 1/2$ we get **Cabinet** projections. Lines perpendicular to the projection plane are foreshortened by half.
- β is typically 30° or 45° .

Perspective Projections

Perspective Projection



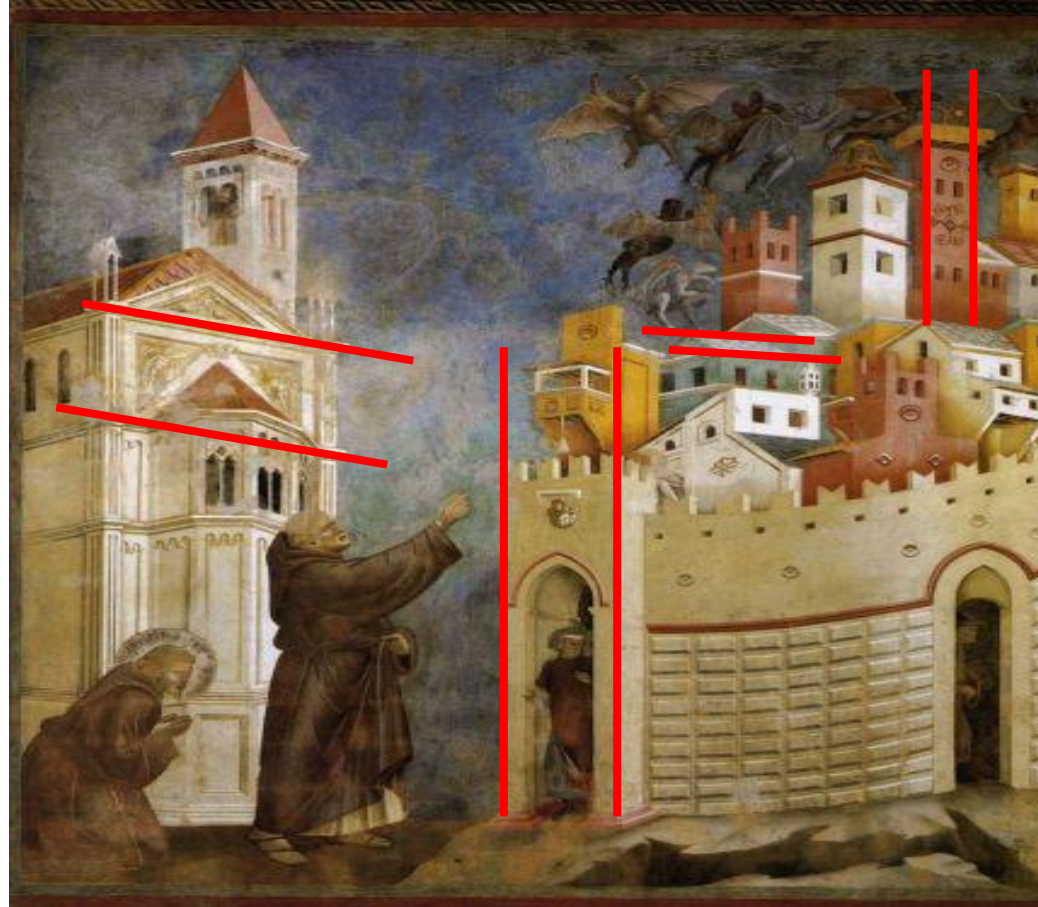
- Projectors converge at a finite centre of projection.
- Parallel lines converge.
- We get non-uniform foreshortening.
- Shape is not preserved.
- We see in perspective – so perspective viewing seems natural and helps in depth perception.

A digression into art



13th century, Arezzo by Giotto

A digression into art



13th century, Arezzo by Giotto

A digression into art



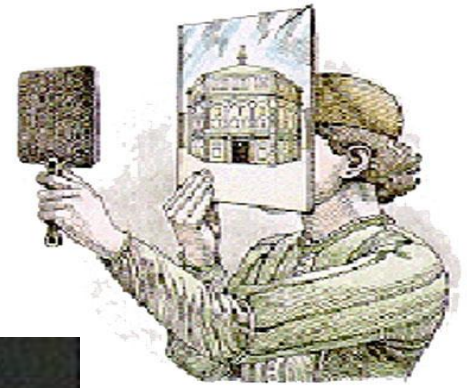
Early 15th century, The Little Garden of Paradise

A digression into art

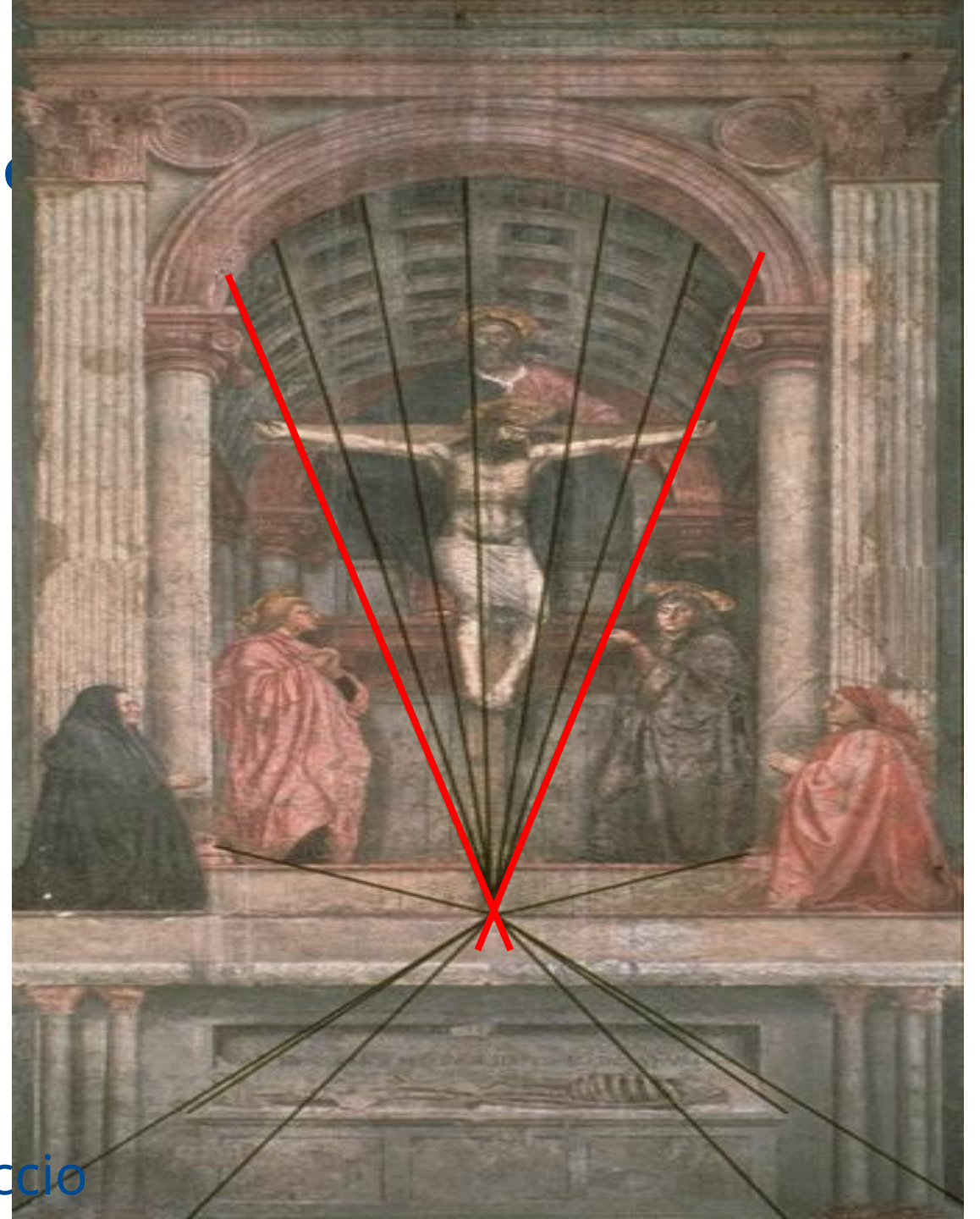


Early 15th century, The Little Garden of Paradise

A digression into art



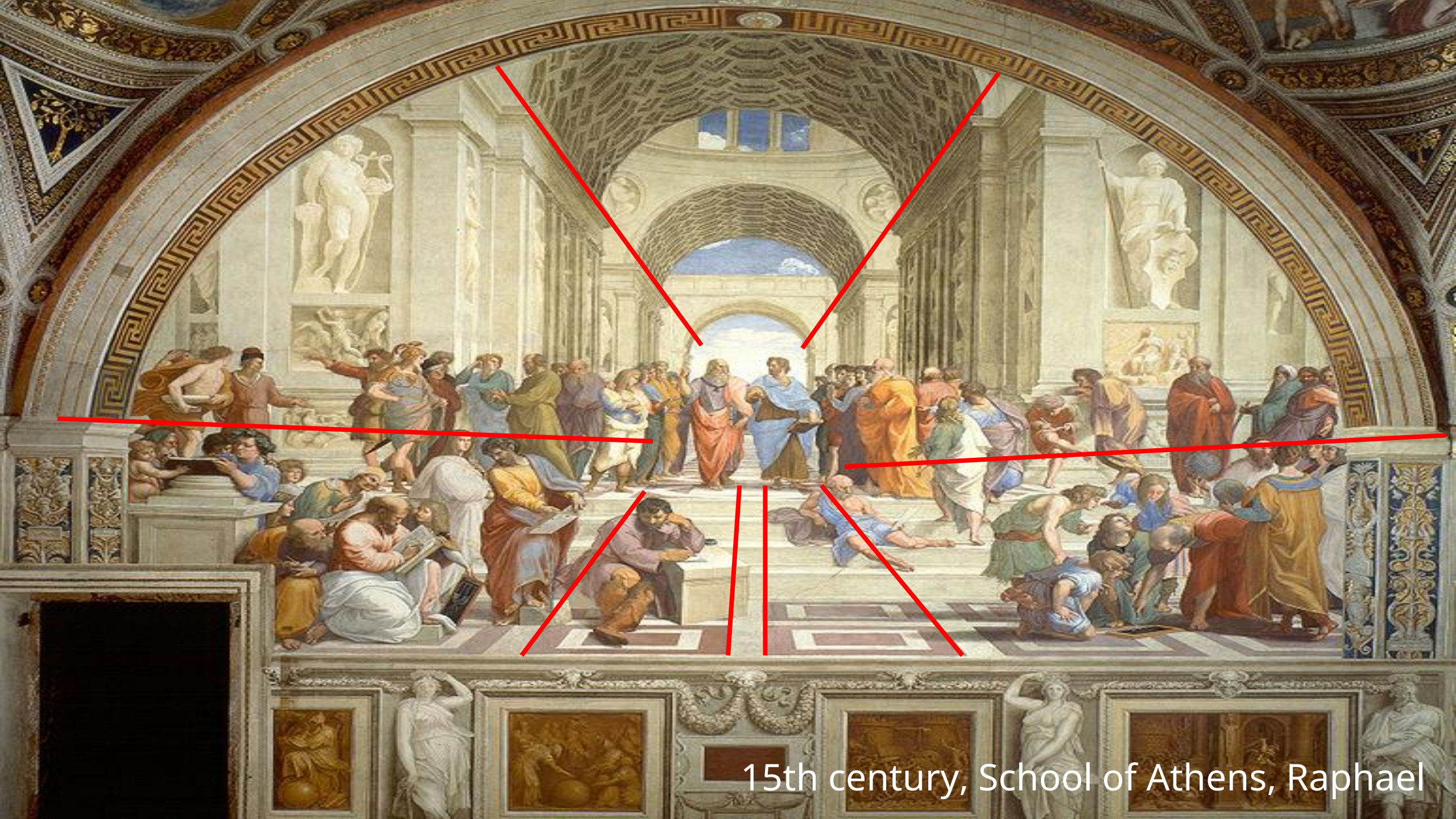
15th century,
The Baptistry
in Florence,
Filippo
Brunelleschi



15th century, Fresco of Holy Trinity, Masaccio



15th century, School of Athens, Raphael

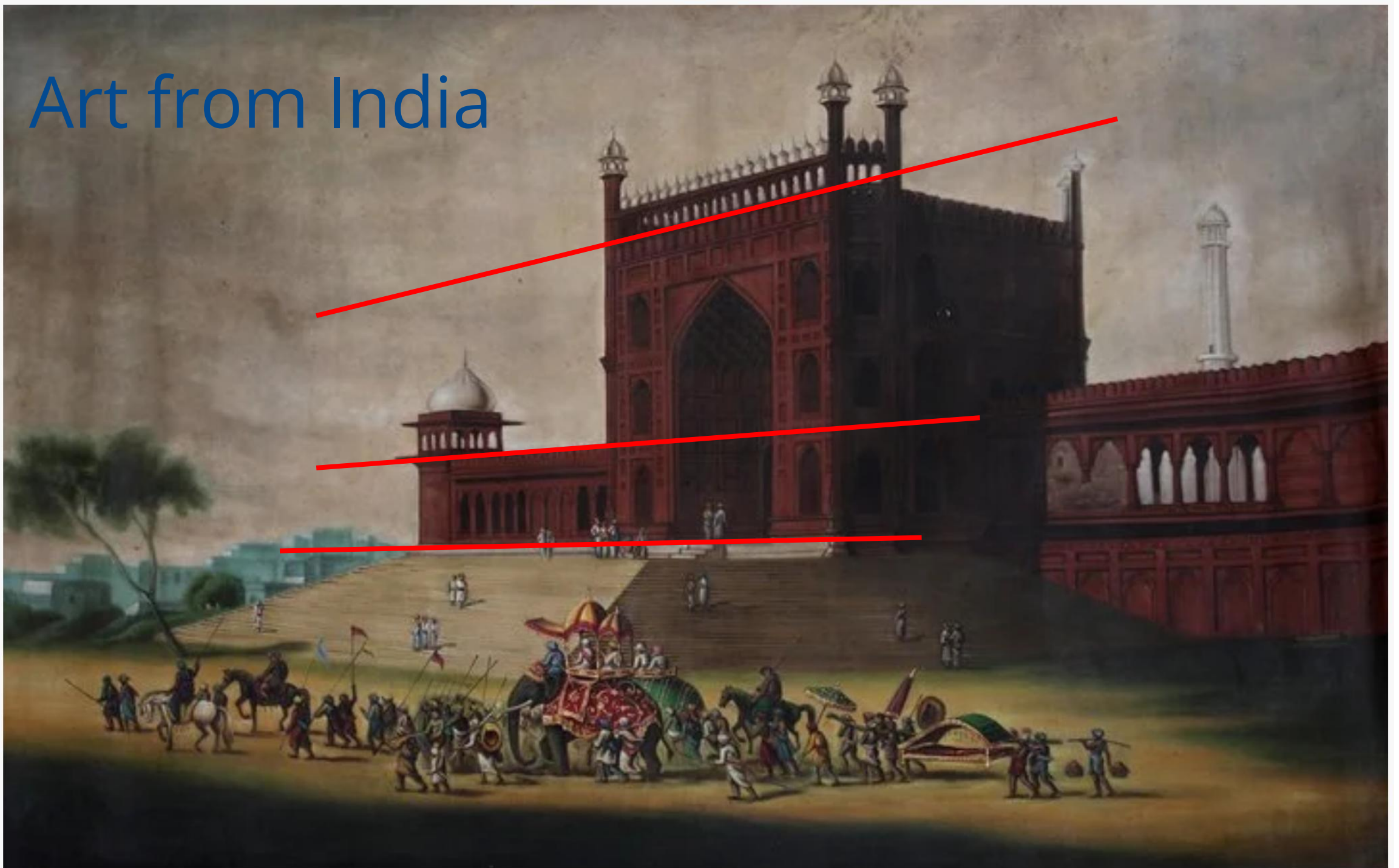


15th century, School of Athens, Raphael

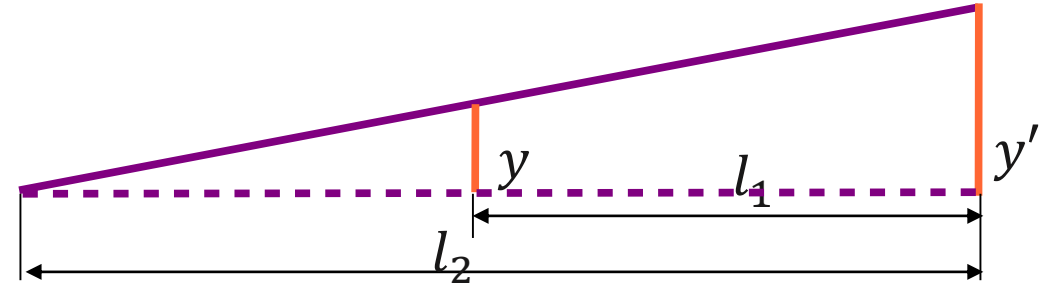
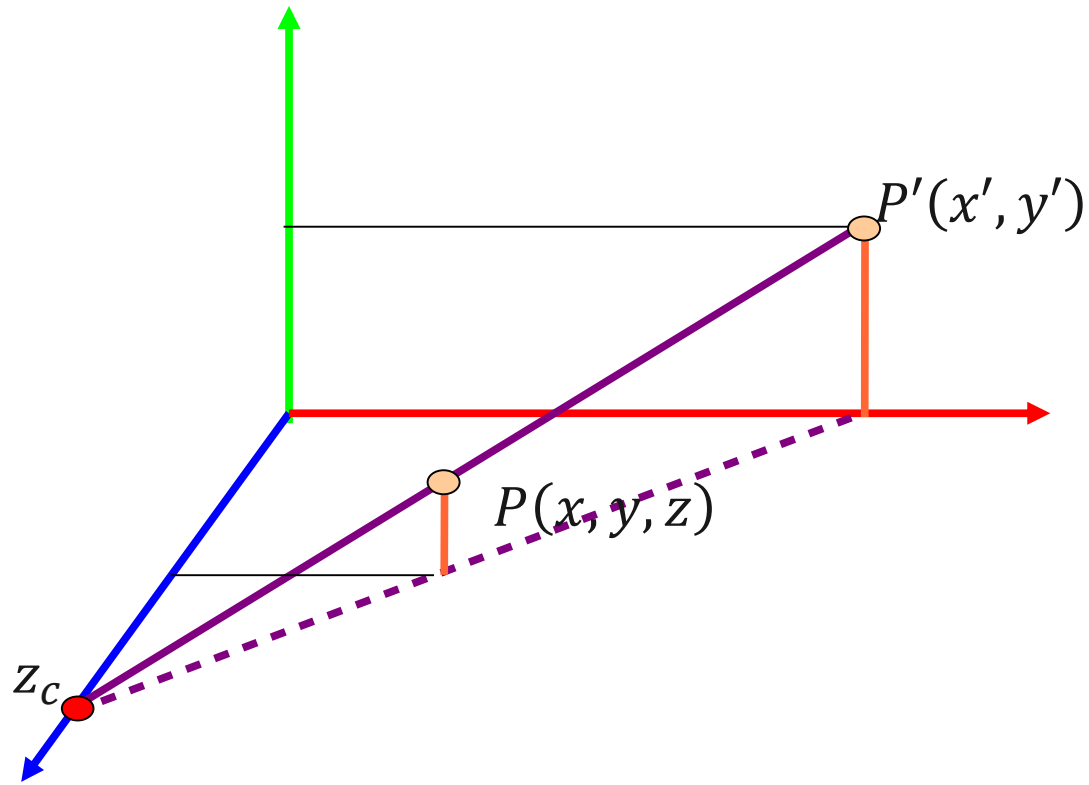
Art from India



Art from India



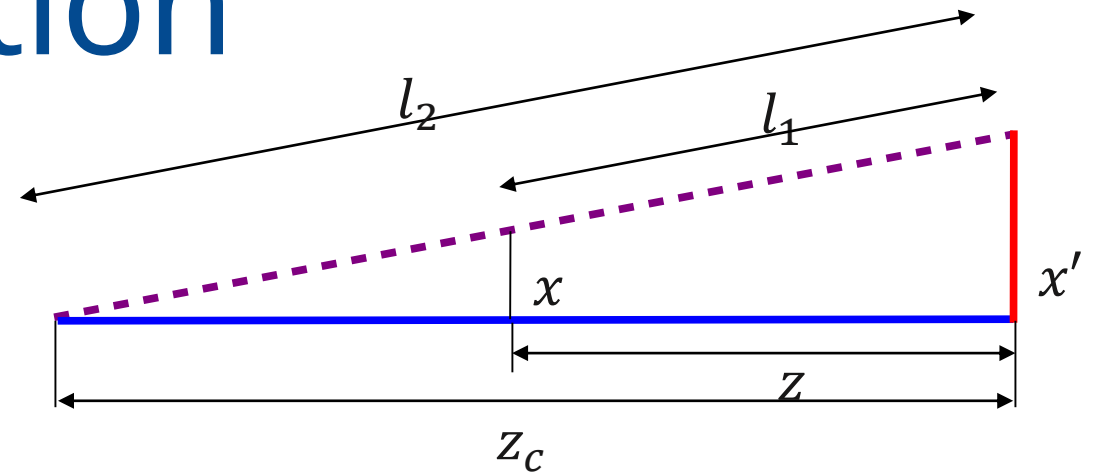
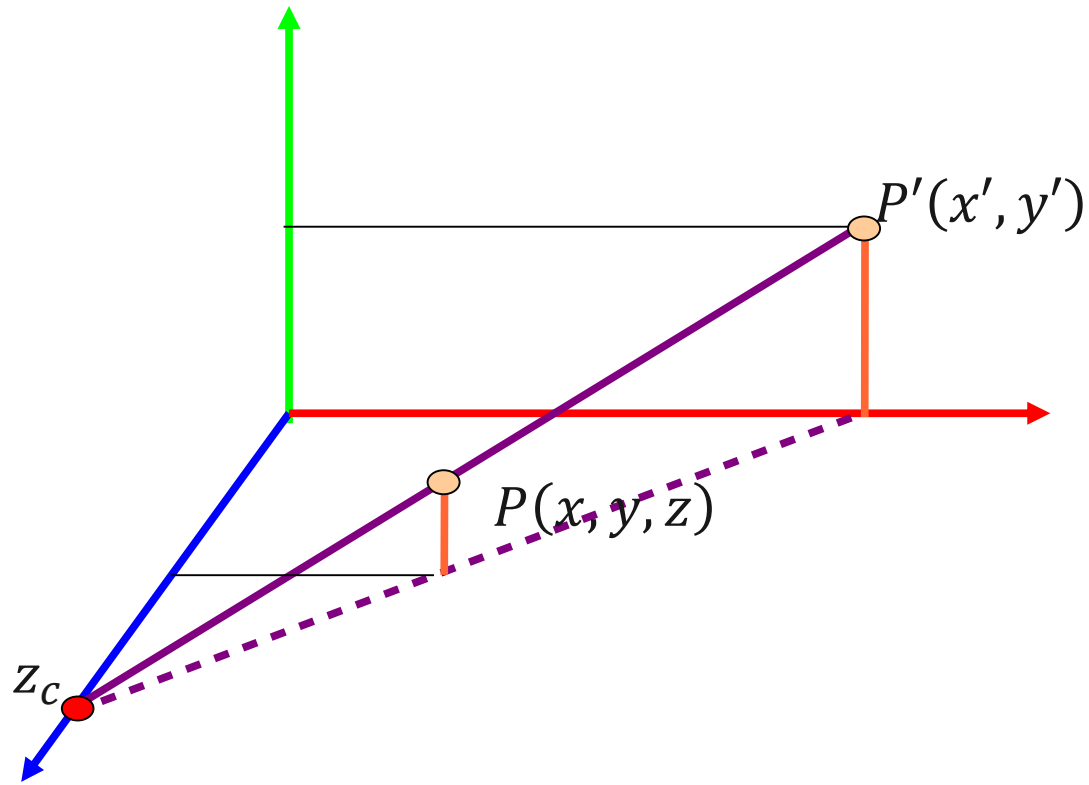
Perspective Projection



$$\frac{y'}{l_2} = \frac{y}{l_2 - l_1} \quad \frac{z_c}{l_2} = \frac{z_c - z}{l_2 - l_1}$$

$$\Rightarrow y' = \frac{y}{1 - \frac{z}{z_c}}$$

Perspective Projection



$$\frac{x'}{l_2} = \frac{x}{l_2 - l_1}$$

$$\frac{Z_c}{l_2} = \frac{Z_c - Z}{l_2 - l_1}$$

$$\Rightarrow x' = \frac{x}{1 - \frac{Z}{Z_c}}$$

Perspective Projection

- First we apply a perspective transform to a point X that takes it to X'

$$X' = P_r \cdot X$$
$$\begin{bmatrix} x' \\ y' \\ z' \\ w' \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & r & 1 \end{bmatrix} \cdot \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix} = \begin{bmatrix} x \\ y \\ z \\ rz + 1 \end{bmatrix}$$
$$x' = \frac{x}{rz + 1}, y' = \frac{y}{rz + 1}, z' = \frac{z}{rz + 1}$$

Perspective Projection

- First we apply a perspective transform to a point X that takes it to X'
- Now we add projection on the $z = 0$ plane.

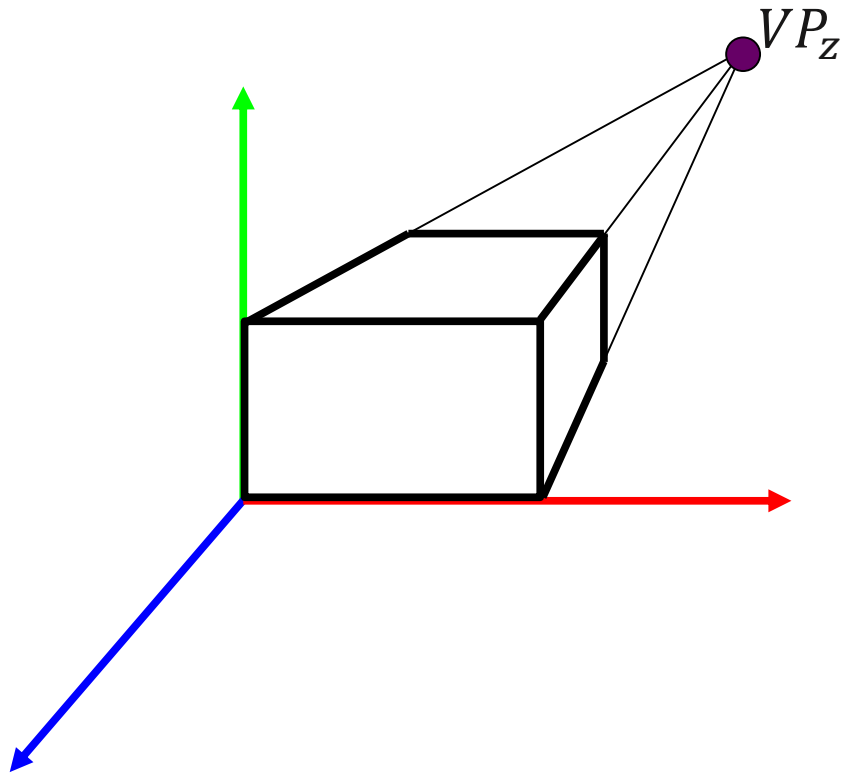
$$X' = P(z = 0) \cdot P_r \cdot X$$
$$\begin{bmatrix} x' \\ y' \\ z' \\ w' \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & r & 1 \end{bmatrix} \cdot \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix} = \begin{bmatrix} x \\ y \\ 0 \\ rz + 1 \end{bmatrix}$$

$$x' = \frac{x}{rz + 1}, y' = \frac{y}{rz + 1}, z' = 0$$

- If $r = \frac{-1}{z_c}$ then we get $x' = \frac{x}{1 - \frac{z}{z_c}}, y' = \frac{y}{1 - \frac{z}{z_c}}$

Types of Perspective Projections

Perspective Projection



- Vanishing point in the z direction.
- Set of lines not parallel to the projection plane converge at a vanishing point.



Homogenous Coordinates

- A point in 3D Euclidean space $P = [x \ y \ z]^T$ is represented by the 4D homogenous point $P_h = [x \ y \ z \ 1]^T$
- A vector (or direction) in 3D Euclidean space $V = [x \ y \ z]^T$ is represented by the 4D homogenous point at infinity in that direction, $V_h = [x \ y \ z \ 0]^T$
- By setting the homogenous coordinate to *zero*, we can treat *points at infinity* (or analogues to directions in 3D Euclidean space) as normal points in homogenous space.

Perspective Projection

- To find the vanishing point along the z direction we apply the perspective transformation to the point at infinity along the z direction.

$$X' = P_r \cdot X$$
$$\begin{bmatrix} x' \\ y' \\ z' \\ w' \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & r & 1 \end{bmatrix} \cdot \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 1 \\ r \end{bmatrix}$$
$$x' = 0, y' = 0, z' = \frac{1}{r}$$

- If $r = -1/z_c$ then we get $z' = -z_c$, i.e., the vanishing point lies an equal distance on the opposite side of the projection plane as the center of projection.

Perspective Projection

- To find the vanishing point along the z direction we apply the perspective transformation to the point at infinity along the z direction.

$$X' = P_r \cdot X$$
$$\begin{bmatrix} x' \\ y' \\ z' \\ w' \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & r & 1 \end{bmatrix} \cdot \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 1 \\ r \end{bmatrix}$$
$$x' = 0, y' = 0, z' = \frac{1}{r}$$

- If $r = -1/z_c$ then we get $z' = -z_c$, i.e., the vanishing point lies an equal distance on the opposite side of the projection plane as the center of projection.

Single point perspective

Perspective Projection

Single point perspective

- Centre of projection (CoP) on x axis

$$X' = P_p \cdot X$$
$$\begin{bmatrix} x' \\ y' \\ z' \\ w' \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ p & 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix} = \begin{bmatrix} x \\ y \\ z \\ px + 1 \end{bmatrix}$$

$$x' = \frac{x}{px + 1}, y' = \frac{y}{px + 1}, z' = \frac{z}{px + 1}$$

- CoP is at $(-1/p, 0, 0, 1)$,VP is at $(1/p, 0, 0, 1)$

Perspective Projection

Single point perspective

- Centre of projection (CoP) on y axis

$$X' = P_q \cdot X$$

$$\begin{bmatrix} x' \\ y' \\ z' \\ w' \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & q & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix} = \begin{bmatrix} x \\ y \\ z \\ qy + 1 \end{bmatrix}$$

$$x' = \frac{x}{qy + 1}, y' = \frac{y}{qy + 1}, z' = \frac{z}{qy + 1}$$

- CoP is at $(0, -1/q, 0, 1)$, VP is at $(0, 1/q, 0, 1)$

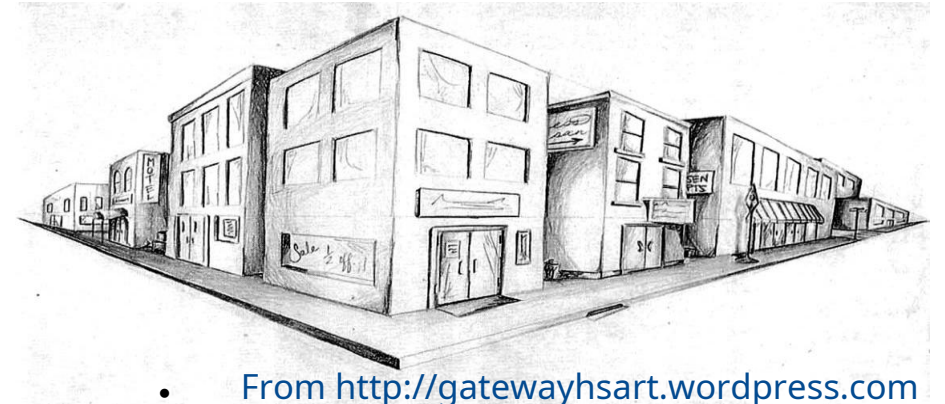
Perspective Projection

Two point perspective

$$P_{pq} = P_p \cdot P_q = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ p & 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & q & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ p & q & 0 & 1 \end{bmatrix}$$

$$x' = \frac{x}{px + qy + 1}, y' = \frac{y}{px + qy + 1}, z' = \frac{z}{px + qy + 1}$$

- Two vanishing points
- Two CoPs?



• From <http://gatewayhsart.wordpress.com>

Perspective Projection

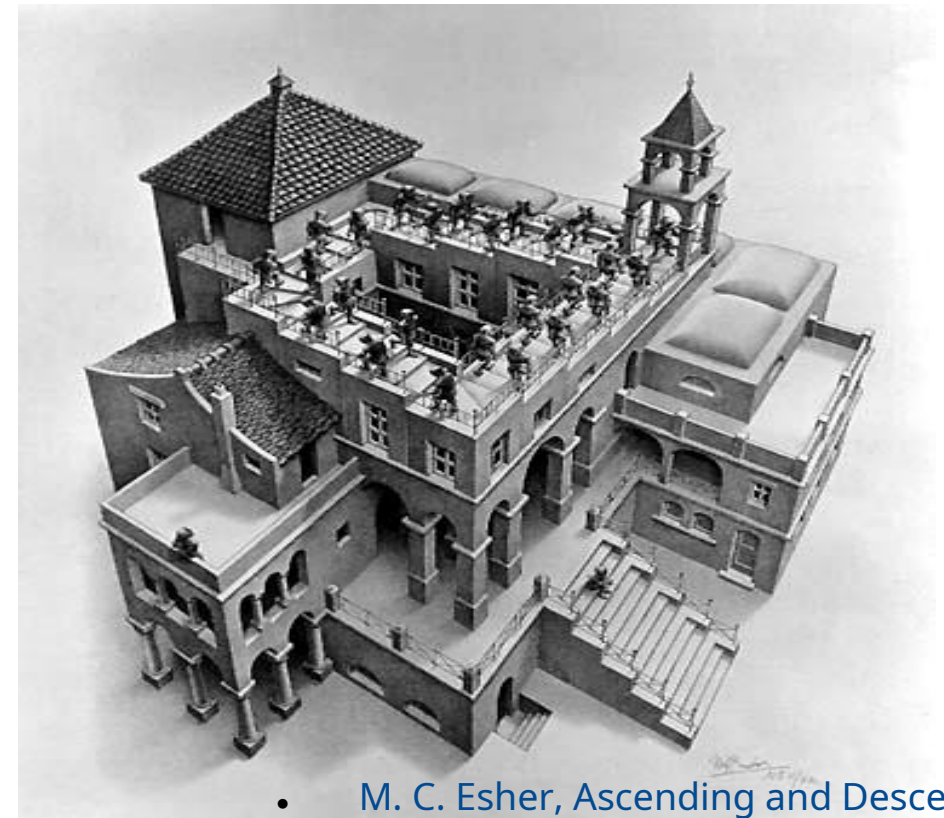
Three point perspective

$$P_{pqr} = P_p \cdot P_q \cdot P_r = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ p & q & r & 1 \end{bmatrix}$$

$$x' = \frac{x}{px + qy + rz + 1}, y' = \frac{y}{px + qy + rz + 1}$$

$$z' = \frac{z}{px + qy + rz + 1}$$

- Three vanishing points
- Three CoPs?



• M. C. Escher, Ascending and Descending

Perspective Projection

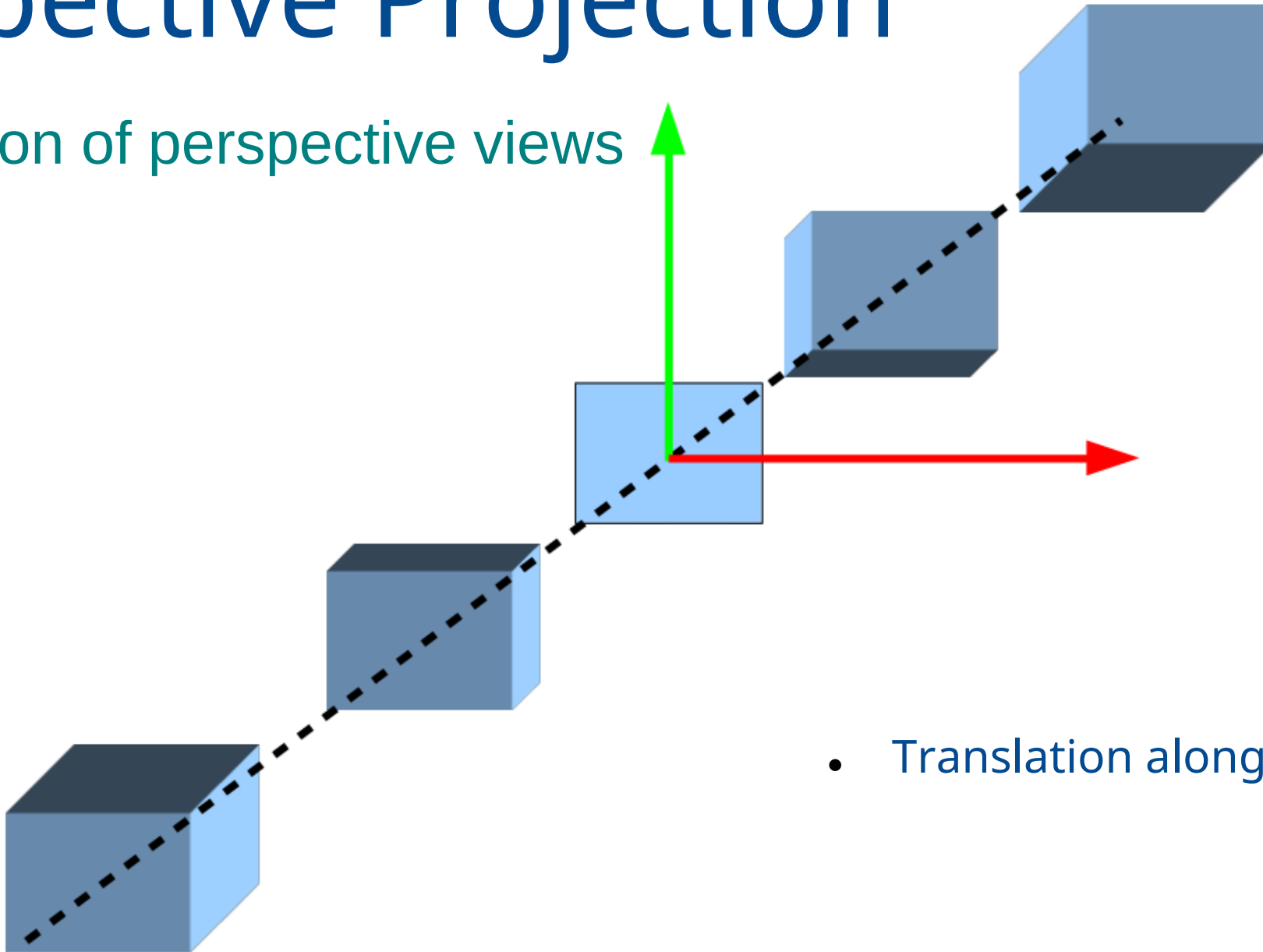
Generation of perspective views

- Transform and then apply single point perspective.
- Let us try to translate, apply a perspective and project to $z=0$

$$T = P_r(z = 0) \cdot T(l, m, n) = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & r & 1 \end{bmatrix} \cdot \begin{bmatrix} 1 & 0 & 0 & l \\ 0 & 1 & 0 & m \\ 0 & 0 & 1 & n \\ 0 & 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & l \\ 0 & 1 & 0 & m \\ 0 & 0 & 0 & 0 \\ 0 & 0 & r & rn + 1 \end{bmatrix}$$

Perspective Projection

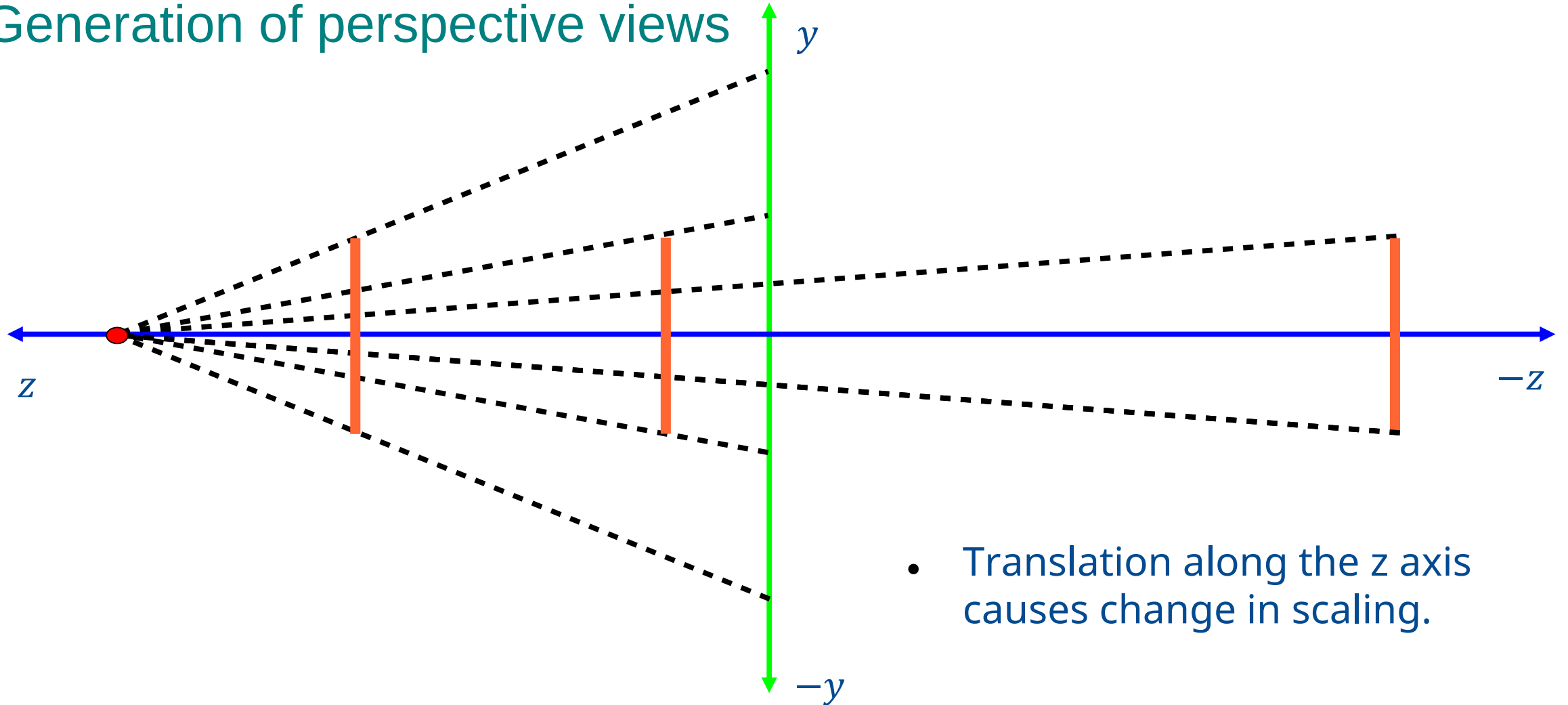
Generation of perspective views



- Translation along $x=y$ line.

Perspective Projection

Generation of perspective views



- Translation along the z axis causes change in scaling.

Perspective Projection

Generation of perspective views

- Rotate about y axis and then apply single point perspective projection.

$$T = P_r(z = 0).R_y(\phi) = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & r & 1 \end{bmatrix} \cdot \begin{bmatrix} \cos\phi & 0 & \sin\phi & 0 \\ 0 & 1 & 0 & 0 \\ -\sin\phi & 0 & \cos\phi & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Perspective Projection

Generation of perspective views

- Rotate about y axis and then apply single point perspective projection.

$$T = P_r(z = 0).R_y(\phi) = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & r & 1 \end{bmatrix} \cdot \begin{bmatrix} \cos\phi & 0 & \sin\phi & 0 \\ 0 & 1 & 0 & 0 \\ -\sin\phi & 0 & \cos\phi & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} \cos\phi & 0 & \sin\phi & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ -r\sin\phi & 0 & r\cos\phi & 1 \end{bmatrix}$$

- We get a two point perspective.

Perspective Projection

Generation of perspective views

- Rotate about y axis, x axis and then apply single point perspective projection.

$$\begin{aligned} T &= P_r(z = 0) \cdot R_x(\theta) \cdot R_y(\phi) \\ &= \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & r & 1 \end{bmatrix} \cdot \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos\theta & -\sin\theta & 0 \\ 0 & \sin\theta & \cos\theta & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} \cos\phi & 0 & \sin\phi & 0 \\ 0 & 1 & 0 & 0 \\ -\sin\phi & 0 & \cos\phi & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \end{aligned}$$

Perspective Projection

Generation of perspective views

- Rotate about y axis, x axis and then apply single point perspective projection.

$$T = P_r(z = 0) \cdot R_x(\theta) \cdot R_y(\phi)$$
$$= \begin{bmatrix} \cos\phi & 0 & \sin\phi & 0 \\ \sin\theta\sin\phi & \cos\theta & -\sin\theta\cos\phi & 0 \\ 0 & 0 & 0 & 0 \\ -r\cos\theta\sin\phi & r\sin\theta & r\cos\theta\cos\phi & 1 \end{bmatrix}$$

- We get a three point perspective.

Taxonomy

